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Inventor(s)	Suh; Yusuhk et al.

Electronic device with antenna modules

Abstract

An antenna module implemented as a multi-layered package includes: a printed circuit board (PCB) having a plurality of layers; and an array antenna portion having a plurality of antenna elements disposed on the PCB. Each of the plurality of antenna elements is configured in a two-patch antenna structure, and a plurality of first patch antennas disposed on a first surface of the PCB are disposed to be spaced apart by a third gap. An area between the plurality of second patch antennas disposed inside the PCB includes a first region in which a plurality of second patch antennas are disposed to be spaced apart by a first gap, and a second region in which the plurality of second patch antennas are disposed to be spaced apart by a second gap.

Inventors: Suh; Yusuhk (Seoul, KR), Lee; Dongik (Seoul, KR), Woo; Seungmin (Seoul, KR)

Applicant: LG ELECTRONICS INC. (Seoul, KR)

Family ID: 1000008818980

Assignee: LG ELECTRONICS INC. (Seoul, KR)

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Primary Examiner: Duong; Dieu Hien T
Attorney, Agent or Firm: LEE, HONG, DEGERMAN, KANG & WAIMEY

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) Pursuant to 35 U.S.C. § 119(a), this application claims the benefit of the earlier filing date and the right of priority to Korean Patent Application No. 10-2023-0081158, filed on Jun. 23, 2023, the contents of which are all incorporated by reference herein in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to an electronic device, and more particularly, to a display device with antenna modules that wirelessly receive data.

BACKGROUND

(3) As image technology changes from analog to digital, development has been made from SD (Standard-Definition) to HD (Hi-Definition) to provide an image closer to a real world. SD supports a resolution of 704×480 and consists of about 350,000 pixels, and HD is divided into HD and Full HD. Between them, Full HD supports a resolution of 1920×1080 and consists of 2 million

pixels to provide a significantly higher quality image compared to SD.

(4) Recent image technology is growing one step further to Ultra High-Definition (UHD) beyond Full HD, and the UHD, which supports high image quality and ultra-high resolution, is spotlighted as a next-generation media environment. The UHD supports 4K (3840×2160) and 8K (7680×4320) resolutions and surround audio of up to 22.2 channels. Compared to the HD, the UHD provides 4 times higher picture quality than the 4K UHD, and the 8K UHD provides 16 times higher image quality than the HD.

(5) In recent years, a wireless display system that wirelessly transmits such a high-resolution image to a display device has emerged.

(6) The wireless display system is a system that transmits and receives A/V data between an A/V transmitting device and an A/V receiving device through a local area network.

(7) The A/V receiving device displays A/V data received from the A/V transmitting device.

(8) An example of the A/V transmitting device may be a transmission box having an antenna module that wirelessly transmits A/V data.

(9) An example of the A/V receiving device may be a display device provided with an antenna module that receives A/V data transmitted from the A/V transmitting device to output the received A/V data.

(10) The display device may include a pair of antenna modules and an IR module located between the pair of antenna modules, and the pair of antenna modules may be disposed to spaced apart from each other on left and right sides thereof.

(11) In the wireless display system, an antenna module of the A/V transmitting device may be located on the left or right side of the display device, and in this case, a pair of antenna modules provided in the display device may receive data transmitted from the antenna module of the A/V transmitting device in a two-stream method, and the display device may output an image.

(12) When the A/V transmitting device is disposed on the left or right side of the display device, one of the pair of antenna modules of the display device cannot receive data because its signal is blocked by the IR module, and the display device operates with one stream.

(13) When operating with one stream, its compression rate must be doubled compared to the case with two streams to transmit and receive data at the same level as in the case of two streams, but when the compression rate is increased, its image quality level may be decreased.

SUMMARY

(14) An aspect of the present disclosure is to provide an electronic device capable of performing wireless communication of A/V data regardless of the location of an A/V transmitting device.

(15) Another aspect of the present disclosure is to perform A/V wireless communication in an optimized manner according to an array antenna disposition structure of an A/V transmitting device and an electronic device.

(16) Still another aspect of the present disclosure is to perform A/V wireless communication in an optimized manner in consideration of the location of an A/V transmitting device and an electronic device, and the polarization characteristics of an array antenna.

(17) Yet still another aspect of the present disclosure is to provide seamless A/V wireless communication even when an obstacle is disposed on a wireless communication path between an A/V transmitting device and an electronic device.

(18) Yet still another aspect of the present disclosure is to implement an antenna module that is capable of transmitting signals over a long distance to a front area of an A/V transmission device and that is also capable of transmitting signals upward.

(19) Yet still another aspect of the present disclosure is to implement an antenna module that is capable of implementing a wider beam coverage in side regions of an A/V transmission device than that in a front or bottom area.

(20) An antenna module implemented as a multi-layered package according to the present disclosure includes: a printed circuit board (PCB) having a plurality of layers; and an array antenna

portion having a plurality of antenna elements disposed on the PCB. Each of the plurality of antenna elements is configured in a two-patch antenna structure, and a plurality of first patch antennas disposed on a first surface of the PCB are disposed to be spaced apart by a third gap. An area between the plurality of second patch antennas disposed inside the PCB includes a first region in which a plurality of second patch antennas are disposed to be spaced apart by a first gap, and a second region in which the plurality of second patch antennas are disposed to be spaced apart by a second gap.

(21) In an embodiment disclosed herein, a first patch antenna of the two patch antennas may be located on a first surface of the PCB, and the first surface may be an outermost surface of the PCB. A second patch antenna of the two patch antennas may be disposed inside the PCB. A part of the first patch antenna and a part of the second patch antenna may be stacked to overlap each other.

(22) In an embodiment disclosed herein, the antenna module may include a radio frequency integrated circuit (RFIC) chip bonded to a second surface of the PCB. The second surface may be another outermost surface of the PCB. The antenna module may include a plurality of signal connection lines configured to connect from the RFIC chip to the array antenna portion. The plurality of signal connection lines may be connected and fed respectively to the second patch antennas of the plurality of antenna elements disposed inside the PCB. A length of the first gap may be shorter than a length of the third gap, and the length of the third gap may be shorter than a length of the second gap.

(23) An electronic device according to an embodiment of the present disclosure may perform wireless communication of A/V data regardless of the location of an A/V transmitting device through first and second antenna structures in which a plurality of array antennas are disposed.

(24) Furthermore, the A/V transmitting device may transmit two streams of data, thereby minimizing video quality deterioration that occurs when increasing a data compression rate.

(25) In addition, since a horizontally polarized antenna and a vertically polarized antenna can be disposed together on one substrate, thereby allowing an antenna module to be compact and providing a high data reception rate.

(26) Moreover, horizontally and vertically polarized signals may be used according to an array antenna disposition structure of the A/V transmitting device and the electronic device, thereby performing A/V wireless communication with reduced mutual interference while increasing a communication capacity.

(27) Besides, horizontally and vertically polarized signals may be used in consideration of the location of the A/V transmitting device and electronic device the polarization characteristics of the array antennas, thereby performing A/V wireless communication with reduced mutual interference while increasing a communication capacity.

(28) In addition, even when an obstacle is disposed on a wireless communication path between the A/V transmitting device and the electronic device, a beamforming direction may be changed and reflected waves may be used, thereby providing seamless A/V wireless communication.

(29) Also, the number of array antennas disposed in a front area of the antenna module of the A/V transmitting device may be greater than the number of antennas in a side region or bottom region. Accordingly, signals can be transmitted over a longer distance in the front area of the antenna module than in the side region or bottom region. Also, an antenna module that has two-dimensional array antennas and is capable of transmitting signals even upward through beamforming can be implemented.

(30) Also, the number of array antennas disposed in side regions of the antenna module of the A/V transmitting device may be greater than the number of antennas in other areas. Accordingly, an antenna module capable of achieving a wider beam coverage in the side regions than that in a front or bottom region can be implemented.

(31) Further scope of applicability of the present disclosure will become apparent from the following detailed description. It should be understood, however, that the detailed description and

specific examples, such as the preferred embodiments of the present disclosure, are given by way of illustration only, since various modifications and alternations within the spirit and scope of the disclosure will be apparent to those skilled in the art.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIGS. 1 and 2 are front and sectional views illustrating a substrate having antenna modules that may be disposed on one side of an electronic device.
- (2) FIG. 3 shows a perspective view of the antenna module of FIG. 2 and an enlarged view of a partial region.
- (3) FIGS. 4A and 4B are front views illustrating the antenna module of FIGS. 1 to 3 for each layer.
- (4) FIG. 5 is an enlarged view of first and sixth layers among a plurality of layers of a PCB of FIG. 4A.
- (5) FIG. 6 is an enlarged view of tenth and twelfth layers among a plurality of layers of a PCB of FIG. 4B.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (6) A description will now be given in detail of specific embodiments of the present disclosure, together with drawings.
- (7) Hereinafter, a description will be given in more detail of embodiments related to the present disclosure, with reference to the accompanying drawings. In general, a suffix such as “module” and “unit” may be used to refer to elements or components. Use of such a suffix herein is merely intended to facilitate description of the specification, and the suffix itself is not intended to give any special meaning or function.
- (8) Hereinafter, an antenna module disposed in an electronic device according to the present disclosure will be described. In this regard, FIGS. 1 and 2 are front and sectional views illustrating a substrate having an antenna module that may be disposed on one side of an electronic device. FIG. 3 shows a perspective view of the antenna module of FIG. 2 and an enlarged view of a partial area.
- (9) (a) of FIG. 1 shows a substrate with an antenna module **1000b** for each area. A substrate **1010a** may include a central region CR and a periphery PE surrounding the central region CR. The periphery PE of the substrate **1010a** may include a first part P1 to a fourth part P4. The first part P1 constitutes a bottom region of the substrate **1010a**, and the second part P2 constitutes one side region of the substrate **1010a**. The third part P3 constitutes another side region of the substrate **1010a**, and the fourth part P4 constitutes a top region of the substrate **1010a**.
- (10) Referring to FIGS. 1 and 2, the antenna module **1000a** may include a substrate **1010a**, a first array antenna **1200a**, a second array antenna **1300a**, a third array antenna **1100a**, and a fourth array antenna **1100b**. Since the third and fourth array antennas **1100a** and **1100b** are disposed on the one side region and the another side region of the substrate **1010a**, they may be referred to as first and second side array antennas.
- (11) (a) of FIG. 2 is a sectional view of the antenna module **1000a**, and (b) of FIG. 2 is a sectional view of the substrate **1010a** shown for each layer. (a) of FIG. 3 is a perspective view illustrating one side region based on the center of the antenna module **1000a**. (b) of FIG. 3 is an enlarged view illustrating an area where some of dipole antennas DA1 to DA4 constituting the second array antenna **1300a** of the antenna module are disposed.
- (12) Referring to FIG. 2, an RFIC **1400a** may be disposed on the first layer La1 of the antenna module **1000a**. A plurality of ground layers may be disposed in an inner region of the antenna module **1000a**. For example, first, second, third, and fourth ground layers GND1, GND2, GND3, and GND4 may be disposed on the second, fourth, fifth, and seventh layers La2, La4, La5, and

La7.

(13) Conductive patterns in the inner region of the antenna module **1000a** may be stacked in a height direction with being spaced apart from one another by a plurality of dielectric layers. For example, the first dielectric layer GND1 may be disposed between the first and second layers La1 and La2, and the second dielectric layer GND2 may be disposed between the third and fourth layers La3 and La4. The third dielectric layer GND3 may be disposed between the fourth and fifth layers La4 and La5, and the fourth dielectric layer GND4 may be disposed between the sixth and seventh layers La6 and La7.

(14) A plurality of coplanar waveguide layers may be disposed on the respective layers in the inner region of the antenna module **1000a**. A first coplanar waveguide layer WG1 in which a plurality of signal connection lines and ground portions are formed may be disposed on the third layer La3. A second coplanar waveguide layer WG2 in which a plurality of signal connection lines and ground portions are formed may be disposed on the sixth layer La6.

(15) It may be considered that a plurality of dielectric layers are disposed on respective layers corresponding to ground layers of the antenna module **1000a**. In this regard, a plurality of dielectric layers may be disposed in the inner region of the antenna module **1000a**. For example, first, second, third, and fourth dielectric layers DL1, DL2, DL3, and DL4 may be disposed on the second, fourth, fifth, and seventh layers La2, La4, La5, and La7.

(16) Referring to FIG. 2, the first layer La1 may be an interface layer IL of the RFIC **1400a**, and the second layer La2 is the first ground layer GND1 having ground and via. The first dielectric layer DL1 may be disposed between the first and second layers La1 and La2, and the second layer La2 may include the first ground layer GND1 and the first dielectric layer DL1.

(17) The third layer La3 is a first coplanar waveguide layer WG1 in which a plurality of conductive patterns and ground portions are formed. The fourth layer La4 is the second ground layer GND2 including ground and via. The second dielectric layer DL2 may be disposed between the third and fourth layers La3 and La4, or the fourth layer La4 may include the second dielectric layer DL2.

(18) The third dielectric layer DL3 may be disposed between the fourth and fifth layers La4 and La5, or the fifth layer La5 may include the third dielectric layer DL3. The fifth layer La5 is the third ground layer GND3 including ground and via. The sixth layer La6 is the second coplanar waveguide layer WG2 in which a plurality of conductive patterns and ground portions are formed. The fourth dielectric layer DL4 may be disposed between the sixth and seventh layers La6 and La7, or the seventh layer La7 may include the fourth dielectric layer DL4.

(19) Referring to FIGS. 1 to 3, a ground wall (GW) **1130** is formed on the periphery PE of the substrate **1010a** and includes vias connecting a plurality of layers. The ground wall (GW) **1130** is disposed in one axial direction and another axial direction between patch antennas PA11 to PA28 to surround each of the patch antennas PA11 to PA28.

(20) The ground wall (GW) **1130** operates as a ground for radiation of the patch antennas PA11 to PA28 and may be referred to as a ground cavity wall. The ground wall (GW) **1130** suppresses side surface radiation and rear surface radiation of the patch antennas PA11 to PA28 having a front side radiation structure, and functions as a reflector to the front surface. In addition, the ground wall (GW) **1130** suppresses rear surface radiation in another side direction of monopole antennas MA1 to MA6 having a side surface radiation structure, and functions as a reflector toward the front side in one side direction. In addition, the ground wall (GW) **1130** suppresses rear surface radiation in a top direction of dipole antennas DA1 to DA10 having a bottom radiation structure, and functions as a reflector in a bottom direction.

(21) The first array antenna **1200a** may further include dummy patches DP11 to DP22 disposed on one side and another side of the patch antennas PA11 to PA28. Among the dummy patches DP11 to DP22, the first dummy patch DP11 is disposed between the first patch antenna PA11 in a first row and the second part P2. The second dummy patch DP12 is disposed between the second patch antenna PA12 in the first row and the second part P2. Among the dummy patches DP11 to DP22,

the third dummy patch DP21 is disposed between the first patch antenna PA21 in a second row and the third part P3. The fourth dummy patch DP22 is disposed between the second patch antenna PA22 in the second row and the third part P3.

(22) The ground wall (GW) 1130 may be formed to surround the dummy patches DP11 to DP22. First patch elements 1210 of the plurality of patch antennas PA11 to PA28 may be connected to feed lines. The dummy patches DP11 to DP22 are not connected to the feed lines. Second patch elements 1220 of the plurality of patch antennas PA11 to PA28 are not connected to the feed lines.

(23) A distance between the ground wall (GW) 1130 and the dummy patches DP11 to DP22, respective sizes thereof, and the like may be implemented within a predetermined range based on a half wavelength of an operating frequency of 60 GHz. Layer positions and sizes of conductive plates CP11 to CP28 corresponding to coupling pads and overlap areas with the patch antennas PA11 to PA28 may be designed in consideration of radiation characteristics and disposition characteristics.

(24) Referring to FIGS. 1 to 3, the antenna module 1000a may further include a millimeter wave transceiver circuitry 1400a.

(25) The substrate 1010a may include a first surface S1, a second surface S2, a periphery PE, and a central region CR. The periphery PE may be formed between the first surface S1 and the second surface S2. The first surface S1 may be opposite to the second surface S2. The substrate 1010a may be implemented as a multi-layer substrate. For example, the substrate 1010a may be implemented as a 12-layer substrate, but is not limited thereto, and may vary depending on applications. The first surface S1 of the substrate 1010a may correspond to a surface of a twelfth layer La12.

(26) The substrate 1010a may have a plurality of side surfaces. Among the plurality of side surfaces, the first surface S1 may be disposed to face a front direction of the antenna module 1000a, and the second surface S2 may be disposed to face a rear direction of the antenna module 1000a. Among the plurality of side surfaces, the third and fourth surfaces S3 and S4 may be disposed to face left and right directions, respectively. Among the plurality of side surfaces, a fifth surface S5 may be configured to face a bottom direction of the antenna module.

(27) The third array antenna 1100a and the fourth array antenna 1100b may be disposed on the second part P2 and the third part P3 of the periphery PE of the substrate 1010a. The third array antenna 1100a and the fourth array antenna 1100b may form beam patterns to side regions of the electronic device. The third array antenna 1100a and the fourth array antenna 1100b may radiate horizontally polarized signals to the side regions of the electronic device.

(28) The third array antenna 1100a may include a plurality of monopole antennas MA1 to MA3 disposed on the second part P2 of the periphery PE of the substrate 1010a. The fourth array antenna 1100b may include the plurality of monopole antennas MA4 to MA6 disposed on the third part P3 of the periphery PE of the substrate 1010a. The third array antenna 1100a and the fourth array antenna 1100b may be implemented with three antenna elements on one side and another side of the periphery PE of the substrate 1010a, respectively. The third array antenna 1100a may be implemented as a 1×3 array antenna on one side of the substrate 1010a, but is not limited thereto. The fourth array antenna 1100b may be implemented as a 1×3 array antenna on another side of the substrate 1010a, but is not limited thereto.

(29) The first array antenna 1200a may be disposed on the first surface S1 of the substrate 1010a. The first array antenna 1200a may form a beam pattern toward the front area of the electronic device. The first array antenna 1200a may radiate a horizontally polarized signal to the front area of the electronic device. The first array antenna 1200a may be implemented as 16 antenna elements on the center region CR of the substrate 1010a.

(30) The first array antenna 1200a may include the plurality of patch antennas PA11 to PA18 and PA21 to PA28 disposed on the first surface S1 of the substrate 1010a. The dummy patches DP11 and DP21 may be disposed on one side of the patch antennas PA11 and P21 to suppress side surface radiation. The dummy patches DP12 and DP22 may be disposed on another side of the

patch antennas **PA11** and **P21** to suppress side surface radiation. The first array antenna **1200a** may be implemented as 16 2×8 array antennas on the center region CR of the substrate **1010a**, but is not limited thereto.

(31) Each patch antenna of the first array antenna **1200a** may include first patch elements **1210** and second patch elements **1220**. The second patch elements **1220** may be stacked in a direction perpendicular to the first patch elements **1210** such that signals of the first patch elements **1210** are coupled. The center of the second patch element **1220** may be offset from the center of the first patch element **1210** in one axial direction.

(32) A second gap between adjacent first patch elements **1221** and **1222** may be larger than a first gap between adjacent second patch elements **1211** and **1212**. To this end, the first patch element **1221** in a first column may be disposed to be offset in the left direction with respect to the second patch element **1211** in the first column. Meanwhile, the first patch element **1222** in a second column may be disposed to be offset in the right direction with respect to the second patch element **1212** in the second column. A current flow direction of a signal applied to the second patch element **1211** in the first column is the left direction, and a current flow direction of a signal applied to the second patch element **1212** in the second column is the right direction. The current flow directions of the signals applied to the second patch elements **1211** and **1212** in the first and second columns are opposite to each other. Accordingly, a phase difference of the signals applied to the second patch elements **1211** and **1212** in the first and second columns is supposed to be 180 degrees so that the current flow directions can be the same. To this end, the RFIC **1400a** may control a phase shifter such that the phase difference between the signals applied to the first patch elements **1211** and **1212** in the first and second columns is 180 degrees.

(33) The second array antenna **1300a** may be disposed on the first part P1 of the periphery PE of the substrate **1010a**. The second array antenna **1300a** may form a beam pattern toward the bottom region of the electronic device. The second array antenna **1300a** may radiate a horizontally polarized signal to the bottom region of the electronic device.

(34) The second array antenna **1300a** may include a plurality of dipole antennas DA1 to DA10 disposed on the first part P1 of the periphery PE of the substrate **1010a**. The second array antenna **1300a** may be implemented as 10 antenna elements on the lower side of the periphery PE of the substrate **1010a**. The second array antenna **1300a** may be implemented as 10 1×10 array antennas on the lower side of the periphery PE of the substrate **1010a**, but is not limited thereto.

(35) The plurality of array antennas may be disposed in an X-axial direction (one axial direction) and a Y-axial direction (another axial direction) of the substrate **1010a**. The third array antenna **1100a** and the fourth array antenna **1100b** may include a plurality of monopole antennas MA1 to MA3 and MA4 to MA6 disposed in the another axial direction. The first array antenna **1200a** may include a plurality of patch antennas PA11 to PA18 and PA21 to PA28 disposed in the one axial direction. The second array antenna **1300a** may include a plurality of dipole antennas DA1 to DA10 disposed in the one axial direction.

(36) The millimeter wave transceiver circuitry **1400a** may be disposed on the second surface S2. The millimeter wave transceiver circuitry **1400a** may be configured to transmit and receive signals at frequencies between 10 GHz and 400 GHz using at least one of the first array antenna **1200a**, the second array antenna **1300a**, and the third and fourth array antennas **1100a** and **1100b**. The millimeter wave transceiver circuitry **1400a** may be configured to transmit and receive signals at frequencies between 10 GHz and 400 GHz using at least one of the plurality of monopole antennas MA1 to MA6, the plurality of patch antennas PA11 to PA18 and PA21 to PA28, and the plurality of dipole antennas DA1 to DA10. The millimeter wave transceiver circuitry **1400a** may be referred to as a radio frequency integrated chip (RFIC).

(37) The number of elements of the first array antenna **1200a** forming the beam pattern toward the front area may be set to be greater than the number of elements of the second array antenna **1300a** forming the beam pattern toward the bottom region. The number of elements of the second array

antenna **1300a** forming the beam pattern toward the bottom region may be set to be greater than the number of elements of the third and fourth array antennas **1100a** and **1100b** forming the beam pattern toward the side regions.

(38) In this regard, 16 pins among 32 pins of the RFIC **1400a** may be connected to the first array antenna **1200a** forming the beam pattern toward the front area. Ten pins of the 32 pins of the RFIC **1400a** may be connected to the second array antenna **1300a** forming the beam pattern toward the bottom region. 6 pins of the 32 pins of the RFIC **1400a** may be connected to the third and fourth array antennas **1100a** and **1100b** forming the beam pattern toward the side regions.

(39) In this regard, the first array antenna **1200a** has the largest number of elements, so it can transmit signals over a long distance to the front area of the electronic device, but has a narrow beam coverage. The narrow beam coverage can be supplemented by changing a beam forming direction to a horizontal direction of the front area. Accordingly, the number of elements of the first array antenna **1200a** may be plural in one axial direction and two in another axial direction. For example, the second array antenna **1300a** may be implemented as 2×8 array antennas. A beam may be formed upward by a predetermined angle from the front direction through a phase difference between signals applied between the antenna elements in the first row and the antenna elements in the second row.

(40) The electronic device needs to perform AV wireless communication with another electronic device disposed in a bottom region of the electronic device. For the AV wireless communication, beamforming may be implemented in units of narrow beam coverage in a horizontal direction, which is the one axial direction, in the bottom region of the electronic device. Meanwhile, it is not necessary to transmit a signal to a bottom region of the electronic device over a long distance. Accordingly, the number of elements of the second array antenna **1300a** may be plural in the one axial direction and one in the another axial direction. For example, the second array antenna **1300a** may be implemented as 1×8, 1×10, or 1×12 array antennas.

(41) Signals may be transferred to the side regions of the electronic device in an indoor radio environment where the electronic device is disposed. It is more important to implement a wide beam coverage for the side regions of the electronic device even without beamforming, than to implement a signal transmission over a long distance. In this regard, since the number of elements of the third and fourth array antennas **1100a** and **1100b** is the smallest, a wide beam coverage to the side regions of the electronic device can be achieved. Accordingly, the number of elements of the third and fourth array antennas **1100a** and **1100b** may be plural in the one axial direction and one in the another axial direction. For example, the third and fourth array antennas **1100a** and **1100b** may be implemented as 1×3 array antennas on one side and another side.

(42) Hereinafter, a disposition structure for each layer of the antenna module according to the present disclosure will be described. In this regard, FIGS. 4A and 4B are front views illustrating the antenna module of FIGS. 1 to 3 for each layer.

(43) Hereinafter, each layer of the antenna module **1000a** will be described in detail with reference to FIGS. 1 to 4B. The antenna module **1000a** may be configured by stacking layers from a first layer La1, on which the transceiver circuitry **1400a** is disposed, to a sixth layer La6, on which the feed lines for the first array antenna **1200a** are located. In addition, the antenna module **1000a** may further include layers from a seventh layer La7, which is a ground layer for the sixth layer La6, to a twelfth layer La12, on which antenna elements of the first array antenna **1200a** are disposed.

(44) The transceiver circuitry **1400a** may be disposed on the first layer La1. The transceiver circuitry **1400a** may have a plurality of pins, and connection lines may be connected to the plurality of pins. The transceiver circuitry **1400a** may be disposed based on a center line of the first layer La1 in one axial direction.

(45) The second layer La2 may include a metal layer on the central region CR, so as to be configured as a first ground layer GND1 for the first layer La1. The monopole antennas MA1 to MA6 of the third and fourth array antennas **1100a** and **1100b** may be disposed on one side region

and another side region of the third layer La3.

(46) The dipole antennas DA1 to DA10 of the second array antenna 1300a may be disposed in a bottom region of the third layer La3.

(47) The fourth layer La4 may include a metal layer on the central region CR, so as to be configured as a second ground layer GND2 for the third layer La3. The first and second feed lines of the third layer La3 are disposed between the first ground layer of the second layer La2 and the second ground layer of the fourth layer La4. Accordingly, the first and second feed lines of the third layer La3 constitute a first coplanar waveguide structure in which ground layers are disposed on an upper layer and a lower layer in a heightwise direction. The metal layers of the first and second ground layers may be partially removed so that the first and second type vias can be vertically connected.

(48) The fifth layer La5 may include a metal layer on the central region CR, so as to be configured as a third ground layer GND3 for the sixth layer La6. On the sixth layer La6, third feed lines for the patch antennas PA11 to PA18 and PA21 to PA28 of the first array antenna 1200a may be disposed. Distances between one end portion and another end portion of the third feed lines may be the same.

(49) The outermost dipole antennas DA1 and DA10 of the third layer La3 may be connected through fourth feed lines of the sixth layer La6.

(50) The seventh layer La7 may include a metal layer on the central region CR, so as to be configured as a fourth ground layer GND4 for the sixth layer La6. The third and fourth feed lines of the sixth layer La6 are disposed between the third ground layer of the fifth layer La5 and the fourth ground layer of the fifth layer La5. Accordingly, the third and fourth feed lines of the sixth layer La6 constitute a second coplanar waveguide structure in which ground layers are disposed on an upper layer and a lower layer in a heightwise direction. The metal layers of the third and fourth ground layers may be partially removed so that the second and third type vias can be vertically connected.

(51) As described above, the second, fourth, fifth, and seventh layers La2, La4, La5, and La7 may configure the first to fourth ground layers GND1 to GND4, respectively. The substrate 1010a may include the first ground layer GND1 for the transceiver circuitry 1400a to the fourth ground layer GND4 for the first array antenna 1200a. The third and fourth array antennas 1100a and 1100b may vertically extend from a layer between the first ground layer GND1 and the second ground layer GND2 to the upper layer of the fourth ground layer GND4.

(52) The first array antenna 1200a may be disposed on the upper layer of the fourth ground layer GND4. The second array antenna 1300a may be disposed on a layer between the first ground layer GND1 and the second ground layer GND2. Accordingly, even if the same horizontal polarization is implemented through the first and second array antennas 1200a and 1300a, mutual interference hardly occurs due to the second to fourth ground layers GND2 to GND4.

(53) In the RFIC 1400a, a length of a feed pattern of the first array antenna 1200a may be configured to be the same for all antenna elements. The length of the feed pattern of the first array antenna 1200a may be determined as the sum of a first length L1a to a fourth length L4a. The length of the feed pattern may be configured to be the same for all the patch antennas PA11 to PA18 and PA21 to PA28 of the third array antenna 1200a. First length L1a to the fourth length L4a may be configured to be the same for all the patch antennas PA11 to PA18 and PA21 to PA28. Accordingly, signals applied from the RFIC 1400a to all of the patch antennas PA11 to PA18 and PA21 to PA28 are in phase, and a beam can be formed toward the center point in the front direction.

(54) First and second via pads VP1 and VP2 may be formed in eighth and ninth layers La8 to La9 to vertically connect the third type vias Vc1 to Vc8 and Vc9 to Vc16. Conductive plates CP11 to CP18 and CP21 to CP28 connected to ends of the third type vias Vc1 to Vc8 and Vc9 to Vc16 may be disposed on a tenth layer La10. The conductive plates CP1 to CP18 and CP21 to CP28 may be referred to as power feeding plates. A first gap G1 between the adjacent conductive plates CP11

and CP12 may be shorter than a second gap G2 between the adjacent conductive plates CP12 and CP13.

(55) A metal layer forming a ground wall GW may be partially disposed on the eleventh layer La11. The conductive plates of the monopole antennas MA1 to MA6 configuring the third and fourth array antennas 1100a and 1100b may be disposed on one side region and another side region of the third layer La3 to the eleventh layer La11.

(56) On the twelfth layer La12, the patch antennas PA11 to PA18 and PA21 to PA28 of the second array antenna 1300a may be disposed. Centers of the patch antennas PA11 to PA18 and PA21 to PA28 may be offset in another axis direction from the conductive plates CP11 to CP18 and CP21 to CP28. A third gap G3 between the adjacent patch antennas PA11 and PA12 may be formed to be longer than the first gap G1 and shorter than the second gap G2.

(57) Hereinafter, a feeding structure for each layer of a first array antenna that performs front surface radiation in an antenna module implemented as a multi-layered antenna package according to the present disclosure will be described in detail. In this regard, FIG. 5 is an enlarged view of first and sixth layers among the plurality of layers of the PCB of FIG. 4A. FIG. 6 is an enlarged view of tenth and twelfth layers among a plurality of layers of a PCB of FIG. 4B.

(58) Referring to FIGS. 1 to 6, signal connection lines (feed lines) of the first layer La1 and the sixth layer La6 may be formed in a coplanar waveguide structure. The signal connection lines of the first layer La1 and the sixth layer La6 may be disposed as feed lines FL on a central portion of the coplanar waveguide structure. In the coplanar waveguide structure of the first layer La1 and the sixth layer La6, ground regions GL and GR may be disposed at one side and another side of the feed lines FL on the central portion. A plurality of ground vias may be disposed in the ground regions GL and GR to be connected to ground regions of other layers. Boundaries on one side and another side of the coplanar waveguide structure may be spaced apart from boundaries of the ground regions GL and GR by predetermined distances.

(59) The RFIC 1400a may be disposed on a central portion of the first layer La1. The plurality of pins of the RFIC 1400a may be connected to feed lines of a first side Sd1 as a top region, feed lines of a second side Sd2 as one side region, feed lines of a third side Sd3 as another side region, and feed lines of a fourth side Sd4 as a bottom region.

(60) End portions Vc1 to Vc8 of first to eighth feed lines F1 to F8 may be disposed in the top region with respect to a central axis of the PCB 1010a. End portions Vc9 to Vc16 of ninth to sixteenth feed lines F9 to F16 may be disposed in the bottom region with respect to the central axis of the PCB 1010a.

(61) The first to third feed lines F1, F2, and F3 of the sixth layer La6 are formed in a structure disposed in the top (left) region of the PCB 1010a. The fourth feed line F4 of the sixth layer La6 is formed in a structure connected from the top region back to the top region via the bottom region. A portion of the fourth feed line F4 is disposed at a position overlapping the inside of the RFIC 1400a.

(62) The ninth to eleventh feed lines F9, F10, and F11 of the sixth layer La6 are formed in a structure disposed in the bottom (left) region of the PCB 1010a. The twelfth feed line F12 of the sixth layer La6 is formed in a structure connected from the top region to the bottom region of the PCB 1010a. One end portion Vx12 of the twelfth feed line F12 is disposed in the top region and another end portion Vc12 is disposed in the bottom region. A portion of the twelfth feed line F12 is disposed at a position overlapping the inside of the RFIC 1400a.

(63) The sixth to eighth feed lines F6, F7, and F8 of the sixth layer La6 are formed in a structure disposed in the top (right) region of the PCB 1010a. The fifth feed line F5 of the sixth layer La6 is formed in a structure connected from the top region back to the top region via the bottom region. A portion of the fifth feed line F5 is disposed at a position overlapping the inside of the RFIC 1400a.

(64) The fourteenth to sixteenth feed lines F14, F15, and F16 of the sixth layer La6 are formed in a structure disposed in the bottom (right) region of the PCB 1010a. The thirteenth feed line F13 of

the sixth layer La6 is formed in a structure connected from the top region to the bottom region of the PCB 1010a. One end portion Vx13 of the thirteenth feed line F13 is disposed in the top region and another end portion Vc13 is disposed in the bottom region. A portion of the thirteenth feed line F13 is disposed at a position overlapping the inside of the RFIC 1400a.

(65) End portions of the feed lines at the first side Sd1 of the first layer La1 may be connected to end portions Vx2, Vx3, Vx4, Vx12, Vx13, Vx5, Vx6, and Vx7 of the feed lines F2, F3, F4, F12, F13, F5, F6, and F7 of the sixth layer La6 through the vertical vias. End portions of the feed lines at the second side Sd2 of the first layer La1 may be connected to end portions Vx1, Vx9, and Vx10, and Vx11 of the feed lines F1, F9, F10, and F11 of the sixth layer La6 through the vertical vias. End portions of the feed lines at the third side Sd3 of the first layer La1 may be connected to end portions Vx8, Vx14, Vx15, and Vx16 of the feed lines F8, F14, F15, and F16 of the sixth layer La6 through the vertical vias.

(66) The feed lines F1 to F16 for all antenna elements constituting the first array antenna 1200a may be formed to have the same length on the sixth layer La6. Ground layers on which vias are formed are disposed at one side and another side of the feed lines F1 to F16. Accordingly, the sixth layer La6 on which the feed lines F1 to F16 are formed is configured as a coplanar waveguide layer.

(67) Center positions of another end portions Vc1 to Vc16 of the feed lines F1 to F16 for all antenna elements constituting the first array antenna 1200a correspond to feeding points for all the antenna elements through the vertical vias. In this regard, FIG. 5 is an enlarged view of the tenth and twelfth layers, on which the first and second patch elements are disposed, among the plurality of layers of FIG. 4B.

(68) The feed lines F1 to F16 for all the antenna elements constituting the first array antenna 1200a may be formed in a symmetrical structure with respect to an Y axis as a vertical axis. The first and eighth feed lines F1 and F8 may be formed in a symmetrical structure with respect to the Y axis. A plurality of regions of the first and eighth feed lines F1 and F8 are formed as straight lines parallel to an X axis. Considering a coordinate difference in the vertical axis between the one end portions Vx1 and Vx8 and the another end portions Vc1 and Vc8 of the first and eighth feed lines F1 and F8, partial regions of end points of the first and eighth feed lines F1 and F8 may be formed with a curved portion and an inclined straight line.

(69) The second and seventh feed lines F2 and F7 may be formed in a symmetrical structure with respect to the Y axis. A plurality of regions of the second and seventh feed lines F2 and F7 are formed as straight lines parallel to the X axis. Considering a coordinate difference in the vertical axis between the one end portions Vx2 and Vx7 and the another end portions Vc2 and Vc7 of the second and seventh lines F2 and F7, partial regions of end points of the second and seventh feed lines F2 and F7 may be formed with a curved portion and an inclined straight line. The first and eighth feed lines F1 and F8 may have the same length and also the second and seventh feed lines F2 and F7 may have the same length.

(70) The third and sixth feed lines F3 and F6 may be formed in a symmetrical structure with respect to the Y axis. The third and sixth feed lines F3 and F6 may include two straight lines parallel to the X axis. A distance between the two straight lines of each of the third and sixth feed lines F3 and F6 may be $\frac{1}{4}$ or more of a wavelength corresponding to an operating frequency, so that mutual interference can be maintained below a predetermined level. The first and eighth feed lines F1 and F8 may have the same length, the second and seventh feed lines F2 and F7 may have the same length, and the third and sixth feed lines F3 and F6 may have the same length.

(71) The fourth and fifth feed lines F4 and F5 may be formed in a symmetrical structure with respect to the Y axis. The fourth and fifth feed lines F4 and F5 may include two straight lines parallel to the Y axis. A distance between the two straight lines of each of the fourth and fifth feed lines F4 and F5 may be $\frac{1}{4}$ or more of a wavelength corresponding to an operating frequency, so that mutual interference can be maintained below a predetermined level. The first and eighth feed lines

F1 and F8, the second and seventh feed lines F2 and F7, the third and sixth feed lines F4 and F5, and the fourth and fifth feed lines F4 and F5 may have the same length, respectively.

(72) The ninth and sixteenth feed lines F9 and F16 may be formed in a symmetrical structure with respect to the Y axis. The ninth and sixteenth feed lines F9 and F16 each may include a straight line parallel to the X axis, a straight line inclined upward, and a straight line inclined downward.

(73) The tenth and fifteenth feed lines F10 and F15 may be formed in a symmetrical structure with respect to the Y axis. The tenth and fifteenth feed lines F10 and F15 each may include a straight line parallel to the X axis, a straight line inclined upward, and two straight lines parallel to the Y axis. A distance between the two straight lines of each of the tenth and fifteenth feed lines F10 and F15 may be $\frac{1}{4}$ or more of a wavelength corresponding to an operating frequency, so that mutual interference can be maintained below a predetermined level. The ninth and sixteenth feed lines F9 and F16 may have the same length and also the tenth and fifteenth feed lines F10 and F15 may have the same length.

(74) The eleventh and fourteenth feed lines F11 and F14 may be formed in a symmetrical structure with respect to the Y axis. The eleventh and fourteenth feed lines F11 and F14 each may include two straight lines parallel to the X axis, and two straight lines parallel to the Y axis. A distance between the two straight lines of each of the eleventh and fourteenth feed lines F11 and F14 may be $\frac{1}{4}$ or more of a wavelength corresponding to an operating frequency, so that mutual interference can be maintained below a predetermined level. The ninth and sixteenth feed lines F9 and F16 may have the same length, the tenth and fifteenth feed lines F10 and F15 may have the same length, and the eleventh and fourteenth feed lines F11 and F14 may have the same length.

(75) The twelfth and thirteenth feed lines F12 and F13 may be formed in a symmetrical structure with respect to the Y axis. The twelfth and thirteenth feed lines F12 and F13 each may include a straight line parallel to the X axis, and a straight line parallel to the Y axis. The ninth and sixteenth feed lines F9 and F16, the tenth and fifteenth feed lines F10 and F15, the eleventh and fourteenth feed lines F11 and F14, and the twelfth and thirteenth feed lines F12 and F13 may have the same length, respectively.

(76) In addition, the first to eighth feed lines F1 to F8 in the top region based on the X axis of the sixth layer La6 and the ninth to sixteenth feed lines F9 to F16 in the bottom region based on the X axis of the sixth layer La6 may all be formed in the same way. This can suppress a beam direction from being changed or beam quality from being degraded due to a phase difference applied to each antenna element, which is caused by a difference in length of the feed lines for each layer.

(77) The tenth layer La10 may include a plurality of first patch elements CP11 to CP18 and CP21 to CP28. Among the plurality of first patch elements, the patch elements CP11 and CP12 adjacent to each other in one axial direction may be spaced apart from each other by a first gap G1. Among the plurality of first patch elements, the patch elements CP12 and CP13 adjacent to each other in the one axial direction may be spaced apart from each other by a second gap G2.

(78) The twelfth layer La12 may include a plurality of second patch elements PA11 to PA18 and PA21 to PA28. Among the plurality of second patch elements PA11 to PA18 and PA21 to PA28, the adjacent patch elements may be disposed to be spaced apart from each other equally by a third gap G3 in the one axial direction.

(79) In this regard, the first patch elements CP11 and CP21 in a first row may be disposed to be offset by a first distance Lx1 in a positive axial direction from a center of a window region WR within the ground wall 1130. Meanwhile, the first patch elements CP12 and CP22 in a second row may be disposed to be offset by the first distance Lx1 in a negative axial direction from the center of the window region WR within the ground wall 1130. The first distance Lx1 may be determined as $(G3-G1)/2$.

(80) The first patch elements CP13, CP15, CP17, CP23, CP25, and CP27 in third, fifth, and seventh rows may be disposed to be offset by the first distance Lx1 in the positive axial direction from the center of the window region WR within the ground wall 1130. On the other hand, the first

patch elements CP14, CP16, CP18, CP23, CP26, and CP28 in fourth, sixth, and eighth rows may be disposed to be offset by the first distance Lx1 in the positive axial direction from the center of the window area WR within the ground wall 1130. The first distance Lx1 may be determined as $(G3-G1)/2$.

(81) Feeding point Vc1 to Vc16 of the feed lines F1 to F16 for all the antenna elements constituting the first array antenna 1200a correspond to feeding positions of the first patch elements CP1 to CP18 and CP21 to CP28. Accordingly, the feeding points Vc1 to Vc16 of the feed lines F1 to F16 are vertically connected through the vertical vias to points that are offset from the centers of the first patch elements CP11 to CP18 and CP21 to CP28. Therefore, the feeding positions of the first patch elements CP11 to CP18 and CP21 to CP28 are defined as the points offset from the centers of the first patch elements CP11 to CP18 and CP21 to CP28 according to the feeding points Vc1 to Vc16 determined for impedance matching.

(82) Center positions of another end portions Vc1 to Vc16 of the feed lines F1 to F16 for all the antenna elements constituting the first array antenna 1200a correspond to center positions of the feeding points for all the antenna elements through the vertical vias.

(83) Hereinafter, an antenna module 1000a implemented as a multi-layered antenna package according to the present disclosure will be described with reference to FIGS. 1 to 6.

(84) Referring to FIGS. 1 to 6, the antenna module 1000 may include a PCB 1010a, a phased array antenna portion 1200a, an RFIC 1400a, and a plurality of signal connection lines SL1 to SL4.

(85) The PCB 1010a may include a plurality of layers. The plurality of layers may include a plurality of metal surfaces and dielectric layers. The plurality of metal surfaces may be stacked with the dielectric layers interposed therebetween to be electrically separated from one another.

(86) The phased array antenna portion 1200a may include a plurality of antenna elements PA11 to PA18 and PA21 to PA28 disposed on the PCB 1010a. Each of the plurality of antenna elements PA11 to PA18 and PA21 to PA28 may be configured to radiate a radio signal. Each of the plurality of antenna elements, as illustrated in (b) of FIG. 2, may have a patch antenna structure including at least two patch antennas 1210 and 1220.

(87) The first patch antenna 1220 of the two patch antennas may be disposed on a first surface S1 of the PCB 1010a, and the first surface S1 may be the outermost surface of the PCB 1010a. Among the two patch antennas, the second patch antenna 1210 may be disposed inside the PCB 1010a. A portion of the first patch antenna 1220 and a portion of the second patch antenna 1210 may be stacked to overlap each other.

(88) The RFIC 1400a may be disposed to be bonded on the second surface S2. The second surface S2 may be another outermost surface of the PCB 1010a.

(89) Each of the plurality of signal connection lines may be configured to connect the RFIC 1400a to the phased array antenna portion 1200a. Radio signals may be transferred between the phased array antenna portion 1200a and the RFIC 1400a by the plurality of signal connection lines SL1 to SL4. Each of the plurality of signal connection lines SL1 to SL4 may be fed by being connected to the second patch antenna 1210 inside the PCB 1010a. Each of the plurality of signal connection lines SL1 to SL4 may be configured not to be directly connected to the first patch antenna 1220 on the first surface S1 of the PCB 1010a. A length of each of the plurality of signal connection lines SL1 to SL4 is defined as a connected length between the RFIC 1400a and the phased array antenna portion 1200a. The plurality of signal connection lines may be formed to have the same length.

(90) The first patch antennas 1220 may be disposed equally at a third gap G3. The second patch antennas 1210 may include a pair of patch antennas spaced apart from each other partially by a first gap G1 and partially by a second gap G2. The first gap G1 may be shorter (narrower) than the third gap G3. The third gap G3 may be shorter (narrower) than the second gap G2.

(91) The plurality of first patch antennas 1220 disposed on the first surface S1 of the PCB 1010a may be disposed to be spaced apart from each other by the third gap G3. An area between the plurality of second patch antennas 1210 disposed inside the PCB 1010a may include a first region

in which the second patch antennas **1210** are spaced apart by the first gap **G1**, and a second region in which the second patch antennas **1210** are spaced apart by the second gap **G2**. The third gap **G3** may be shorter (narrower) than the second gap **G2**.

(92) The antenna module **1000** may further include a ground wall **1130** disposed between the plurality of antenna elements. The second patch antennas **1210** may be disposed to be offset adjacent to one side or another side of the ground wall **1130**.

(93) The two-patch antenna structure may be disposed in an area having a plurality of ground regions. The two-patch antenna structure may be disposed in a first ground region **1100g**. The two-patch antenna structure may be disposed in an area including a first horizontal ground region **GH1**, a second horizontal ground region **GH2**, and vertical ground walls **GV1** to **GV4**, **GV12** to **GV34**.

(94) The first horizontal ground region **GH1** may be formed on the first surface **S1**. The second horizontal ground region **GH2** may be formed inside the PCB **1010a**. The second horizontal ground region **GH2** may be defined in the sixth layer **La6** or the seventh layer **La7** of the PCB **1010a**. The vertical ground walls **GV1** to **GV4**, **GV12** to **GV34** may be formed to vertically connect the first horizontal ground region **GH1** and the second horizontal ground region **GH2**.

(95) A plurality of connection lines **SL1** to **SL4** may be disposed between the second patch antennas **1210** and the RFIC chip **1400a** inside the PCB **1010a**. A length of each of the plurality of signal connection lines **SL1** to **SL4** is defined as a connected length between the RFIC **1400a** and the phased array antenna portion **1200a**.

(96) Each of the plurality of signal connection lines **SL1** to **SL4** may include a first part **SL1**, a second part **SL2**, a third part **SL3**, and a fourth part **SL4**. The first part **SL1** may be defined on the second surface **S2** of the PCB **1010a**. The third part **SL3** may be formed inside the PCB **1010a** to configure a coplanar waveguide structure. The second part **SL1** may be defined to electrically connect the first part **SL3** and the third part **SL2**. The plurality of signal connection lines **SL1** to **SL4** may have the same length.

(97) The antenna module **1000a** may include a plurality of ground regions. The antenna module **1000a** may include a first ground region **1300g**, a second ground region **1200a**, and a third ground region **1300g**. The two-patch antenna structure and the dummy patches **DP11**, **DP12**, **DP21**, and **DP22** may be disposed in the first ground region **1100g**.

(98) The second ground region **1200g** that is configured as a metal surface may be defined between the coplanar waveguide structure **WG2** and the second patch antenna **1210**. The third ground region **1300g** that is configured as a metal surface may be defined between the coplanar waveguide structure **WG1** and the RFIC chip **1400a**.

(99) The dummy patches **DP11**, **DP12**, **DP21**, and **DP22** which are spaced apart the third gap **G3** from the outermost first patch antennas **PA11**, **PA18**, **PA21**, and **PA28** of the first path antennas **1220** may be disposed on the first surface **S1** of the PCB **1010a**. The dummy patches **DP11**, **DP12**, **DP21**, and **DP22** may also be disposed in the area including the first horizontal ground region **GH1**, the second horizontal ground region **GH2**, and the vertical ground walls **GV1** to **GV4**, **GV12** to **GV34**.

(100) The first horizontal ground region **GH1** may be formed on the first surface **S1**. The second horizontal ground region **GH2** may be formed inside the PCB **1010a**. The second horizontal ground region **GH2** may be defined in the sixth layer **La6** or the seventh layer **La7** of the PCB **1010a**. The vertical ground walls **GV1** to **GV4**, **GV12** to **GV34** may be formed to vertically connect the first horizontal ground region **GH1** and the second horizontal ground region **GH2**.

(101) The phased array antenna portion **1200a** may include a first array antenna portion **1250a** and a second array antenna portion **1300a**. In the first array antenna portion **1250a**, eight first patch antennas **PA11** to **PA18** and eight second patch antennas **CP11** to **CP18** may be disposed in a first column in the Y-axial direction. In the second array antenna portion **1250b**, eight first patch antennas **PA21** to **PA28** and eight second patch antennas **CP21** to **CP28** may be disposed in a second column in the Y-axial direction. The phased array antenna portion **1200a** may be

implemented as a 2×8 array antenna.

(102) The first, third, fifth, and seventh antenna elements of the first array antenna portion **1250a** may be disposed such that the second patch antennas CP11, CP13, CP15, and CP17 are offset from the first patch antennas PA11, PA13, PA15, and PA17 in a positive Y-axial direction. The second, fourth, sixth, and eighth antenna elements of the first array antenna portion **1250a** may be disposed such that the second patch antennas CP12, CP14, CP16, and CP18 are offset from the first patch antennas PA12, PA14, PA16, and PA18 in a negative Y-axial direction.

(103) The second patch antennas CP11, CP13, CP15, and CP17 corresponding to the first, third, fifth, and seventh antenna elements may be disposed adjacent to another side (right side) of the ground wall **1130**. The second patch antennas CP12, CP14, CP16, and CP18 corresponding to the second, fourth, sixth, and eighth antenna elements may be disposed adjacent to one side (left side) of the ground wall **1130**.

(104) The first, third, fifth, and seventh antenna elements of the second array antenna portion **1250b** may be disposed such that the second patch antennas CP21, CP23, CP25, and CP27 are offset from the first patch antennas PA21, PA23, PA25, and PA27 in the positive Y-axial direction. The second, fourth, sixth, and eighth antenna elements of the second array antenna portion **1250b** may be disposed such that the second patch antennas CP22, CP24, CP26, and CP28 are offset from the first patch antennas PA22, PA24, PA26, and PA28 in the negative Y-axial direction.

(105) The second patch antennas CP21, CP23, CP25, and CP27 corresponding to the first, third, fifth, and seventh antenna elements may be disposed adjacent to another side (right side) of the ground wall **1130**. The second patch antennas CP22, CP24, CP26, and CP28 corresponding to the second, fourth, sixth, and eighth antenna elements may be disposed adjacent to one side (left side) of the ground wall **1130**.

(106) A first feeding point Vc1 of the first antenna element and a second feeding point Vc2 of the second antenna element may be spaced apart from each other by the first gap G1. The second feeding point Vc2 of the second antenna element and a third feeding point Vc3 of the third antenna element may be spaced apart from each other by the second gap G2.

(107) The first feeding point Vc1 connected to the second patch antenna CP11 of the first antenna element may be offset from the center of the second patch antenna CP11 in the positive Y-axial direction. The second feeding point Vc2 connected to the second patch antenna CP12 of the second antenna element may be offset from the center of the second patch antenna CP12 in the negative Y-axial direction. The third feeding point Vc3 connected to the second patch antenna CP13 of the third antenna element may be offset from the center of the second patch antenna CP13 in the positive Y-axial direction.

(108) A direction of current applied to the first, third, fifth, and seventh antenna elements disposed to be offset in the positive Y-axial direction may be formed opposite to a direction of current applied to the second, fourth, sixth, and eighth antenna elements disposed to be offset in the negative Y-axial direction. In order to form a direction of beam pattern of an array antenna toward a center of a front area, the directions of currents applied to all the antenna elements should be the same. Accordingly, the RFIC **1400a** may control a phase shifter to create a phase difference of 180 degrees between signals applied to the first, third, fifth, and seventh antenna elements and signals applied to the second, fourth, sixth, and eighth antenna elements.

(109) The RFIC **1400a** may be disposed on the first layer La1 of the PCB **1010**. The plurality of ground layers may be disposed inside the PCB **1010**. In this regard, the first to fourth ground layers GND1, GND2, GND3, and GND4 may be disposed on the second, fourth, fifth, and seventh layers La2, La4, La5, and La7 of the PCB **1010**.

(110) Each of the plurality of signal connection lines SL1 to SL4 may include a first part SL1, a second part SL2, a third part SL3, and a fourth part SL4. The first parts SL1 of the plurality of signal connection lines may be disposed on the first layer La1 of the PCB **1010**. The second parts SL2 of the signal connection lines may correspond to first vertical vias connecting the first layer

La1 and the sixth layer La6 of the PCB 1010. The third parts SL3 of the plurality of signal connection lines may be disposed on the sixth layer La6 of the PCB 1010. The fourth parts SL4 of the signal connection lines may correspond to second vertical vias that connect the second patch antennas CP11 to CP18, CP21 to CP28 of the sixth layer La6 and the tenth layer La10 of the PCB 1010.

(111) The first parts SL1 may have the same length for the plurality of antenna elements. The second parts SL2 may have the same length for the plurality of antenna elements. The third parts SL3 may have the same length for the plurality of antenna elements. The fourth parts SL4 may have the same length for the plurality of antenna elements.

(112) The first patch antennas PA11 to PA18, PA21 to PA28 and the second patch antennas CP11 to CP18, CP21 to CP28 may be disposed on the twelfth layer La12 and the tenth layer La10 of the PCB 1010, respectively. The fourth parts SL4 of the signal connection lines may vertically connect the second patch antennas CP11 to CP18, CP21 to CP28 of the sixth layer La6 and the tenth layer La10 of the PCB 1010.

(113) The first vertical vias may be disposed on a first side surface Sd1 as an upper area of the RFIC 1400a, a second side surface Sd2 as one side area, and a fourth side surface Sd4 as another side area. The second vertical vias may include a pair of via groups that are spaced apart from each other partially by the first gap G1 and partially by the second gap G2.

(114) The first, third, fifth, and seventh antenna elements of the first array antenna portion 1250a may be disposed such that connection points Vc1, Vc3, Vc5, and Vc7 of the second vertical vias with respect to the first patch antennas PA11, PA13, PA15, and PA17 are offset in the positive Y-axial direction. The second, fourth, sixth, and eighth antenna elements of the first array antenna portion 1250a may be disposed such that connection points Vc2, Vc4, Vc6, and Vc8 of the second vertical vias with respect to the first patch antennas PA12, PA14, PA16, and PA18 are offset in the negative Y-axial direction.

(115) The connection points Vc1, Vc3, Vc5, and Vc7 of the second vertical vias corresponding to the first, third, fifth, and seventh antenna elements may be disposed adjacent to another side (right side) of the ground wall 1130. The connection points Vc2, Vc4, Vc6, and Vc8 of the second vertical vias corresponding to the second, fourth, sixth, and eighth antenna elements may be disposed adjacent to one side (left side) of the ground wall 1130.

(116) The first, third, fifth, and seventh antenna elements of the second array antenna portion 1250b may be disposed such that connection points Vc9, Vc11, Vc13, and Vc15 of the second vertical vias with respect to the first patch antennas PA21, PA23, PA25, and PA27 are offset in the positive Y-axial direction. The second, fourth, sixth, and eighth antenna elements of the second array antenna portion 1250b may be disposed such that connection points Vc10, Vc12, Vc14, and Vc16 of the second vertical vias with respect to the first patch antennas PA22, PA24, PA26, and PA28 are offset in the negative Y-axial direction.

(117) The connection points Vc9, Vc11, Vc13, and Vc15 of the second vertical vias corresponding to the first, third, fifth, and seventh antenna elements may be disposed adjacent to another side (right side) of the ground wall 1130. The connection points Vc10, Vc12, Vc14, and Vc16 of the second vertical vias corresponding to the second, fourth, sixth, and eighth antenna elements may be disposed adjacent to one side (left side) of the ground wall 1130.

(118) The foregoing description has been given of the antenna module implemented as the multi-layered package including the array antennas each having patch antennas of the offset structure according to one aspect of the present disclosure. Hereinafter, a description will be given of a disposition structure of feed lines for each layer in a multi-layered antenna package including a plurality of array antennas having a plurality of coplanar waveguide structures according to another aspect of the present disclosure. In this regard, an antenna module implemented as a multi-layered antenna package including a plurality of array antennas having a plurality of coplanar waveguide structures will be described with reference to FIGS. 1 to 16.

(119) The antenna module **1000a** includes a transceiver circuitry **1400a**, first resonating elements **1300a**, second resonating elements **1200a**, and a plurality of signal connection lines SL1 to SL4. The antenna module **1000a** may further include a first coplanar waveguide WG1 and a second coplanar waveguide WG2. The first resonating elements **1300a** may correspond to the second array antenna **1300a**. The second resonating elements **1200a** may correspond to the first array antenna **1200a** and the phased array antenna portion **1200a**.

(120) The first coplanar waveguide WG1 may be configured to convey first signals at a frequency of 10 GHz or higher between the transceiver circuitry **1400a** and the first resonating element **1300a**. The second coplanar waveguide WG2 may be configured to convey second signals at the frequency of 10 GHz or higher between the transceiver circuitry **1400a** and the second resonating elements **1200a**.

(121) The first coplanar waveguide WG1 may be interposed between the second coplanar waveguide WG2 and the transceiver circuitry **1400a**. The second coplanar waveguide WG2 may be interposed between the first coplanar waveguide WG1 and the second resonating elements **1200a**.

(122) The first resonating elements **1300a** may be interposed between the second coplanar waveguide WG2 and the transceiver circuitry **1400a**. The second resonating elements **1200a** may be disposed on an opposite side of the transceiver circuitry **1400a**. The second resonating elements **1200a** may configure the array antenna.

(123) Each of the second resonating elements **1200a** may include at least two patch antenna layers configured to radiate radio signals. Each of the second resonating elements **1200a** may have a two-patch antenna structure. The first patch antenna **1220** of the two patch antennas may be disposed on the first surface S1 of the PCB **1010a**, and the first surface S1 may be the outermost surface of the PCB **1010a**. The second patch antenna **1210** of the two patch antennas may be disposed inside the PCB **1010a**. A portion of the first patch antenna **1220** and a portion of the second patch antenna **1210** may be stacked to overlap each other.

(124) The plurality of signal connection lines SL1 to SL4 may be configured to connect the transceiver circuitry **1400a** to the phased array antenna portion **1200a**. Each of the plurality of signal connection lines SL1 to SL4 may be fed by being connected to the second patch antennas **1210** disposed inside the PCB **1010a**. Each of the plurality of signal connection lines SL1 to SL4 may be configured not to be directly connected to the first patch antennas **1220** on the first surface S1 of the PCB **1010a**.

(125) A length of each of the plurality of signal connection lines SL1 to SL4 may be defined as a connected length between the transceiver circuitry **1400a** and the phased array antenna portion **1200a**. The plurality of signal connection lines SL1 to SL4 may have the same length for each of the second resonating elements **1200a**. The plurality of signal connection lines SL1 to SL4 may be disposed between the phased array antenna portion **1200a** inside the PCB **1010a** and the transceiver circuitry **1400a**.

(126) Each of the plurality of signal connection lines may include the first part SL1 to the fourth part SL4. The first part SL1 may be disposed on the second surface S2 of the PCB **1010a**. The third part SL3 may be disposed on the second coplanar waveguide WG2. The second part S2 may be configured to electrically connect the first part SL1 and the third part SL3. The fourth part SL4 may be configured to electrically connect the third part SL3 and one of the second patch antennas **1210**.

(127) The second coplanar waveguide WG2 may be disposed on a conductive plate between two ground conductive plates GND3 and GND4 inside the PCB **1010a**. The conductive plate of the second coplanar waveguide WG2 may include signal connection lines SL3 and ground portions. The signal connection lines and ground portions may be disposed on the coplanar conductive plate between the two ground conductive plates GND3 and GND4 inside the PCB **1010a**.

(128) The PCB **1010a** may include first to sixth layers La1 to La6. The first parts SL1 of the plurality of signal connection lines may be disposed on the first layer La1 of the PCB **1010a**. The second parts SL2 of the plurality of signal connection lines may be first vertical vias formed from

the first layer La1 to the sixth layer La6 of the PCB 1010a. The third parts SL3 of the plurality of signal connection lines may be disposed on the sixth layer La6 of the PCB 1010a. The fourth parts SL4 of the plurality of signal connection lines may be second vertical vias formed from the sixth layer La6 to the second patch antenna 1220.

(129) For the second resonating elements 1200a, the first parts SL1 on the first layer La1 may have the same length. For the second resonating elements 1200a, the third parts SL3 on the sixth layer La6 may have the same length.

(130) For the second resonating elements 1200a, the first vertical vias corresponding to the second parts SL2 may have the same length (height). For the second resonating elements 1200a, 1200b, the second vertical vias corresponding to the fourth parts SL4 may have the same height.

(131) The feed lines of the first layer La1 may be connected to the feed lines F1 to F16 of the sixth layer La1 through the first vertical vias passing through the first and second ground layers GND1 and GND2. The feed lines F1 to F16 of the sixth layer La1 may be connected to the second patch antennas 1210. All of the feed lines F1 to F16 of the sixth layer La1 may be formed to have the same length.

(132) The first patch antennas 1220 may be disposed equally at a third gap G3. The second patch antennas 1210 may include a pair of patch antennas spaced apart from each other partially by a first gap G1 and partially by a second gap G2. The first gap G1 may be shorter (narrower) than the third gap G3. The third gap G3 may be shorter (narrower) than the second gap G2.

(133) The plurality of first patch antennas 1220 disposed on the first surface S1 of the PCB 1010a may be disposed to be spaced apart from each other by the third gap G3. An area between the plurality of second patch antennas 1210 disposed inside the PCB 1010a may include a first region in which the second patch antennas 1210 are spaced apart by the first gap G1, and a second region in which the second patch antennas 1210 are spaced apart by the second gap G2. The third gap G3 may be shorter (narrower) than the second gap G2.

(134) The antenna module 1000a may further include a ground wall 1130 disposed between the second resonating elements 1200a. The second patch antennas 1210 may be disposed to be offset adjacent to one side or another side of the ground wall 1130.

(135) The two-patch antenna structure may be disposed in an area having a plurality of ground regions. The two-patch antenna structure may be disposed in a first ground region 1100g. The two-patch antenna structure may be disposed in an area including a first horizontal ground region GH1, a second horizontal ground region GH2, and vertical ground walls GV1 to GV4, GV12 to GV34.

(136) The first horizontal ground region GH1 may be formed on the first surface S1. The second horizontal ground region GH2 may be formed in the PCB 1010a. The second horizontal ground region GH2 may be defined in the sixth layer La6 or the seventh layer La7 of the PCB 1010a. The vertical ground walls GV1 to GV4, GV12 to GV34 may be formed to vertically connect the first horizontal ground region GH1 and the second horizontal ground region GH2.

(137) A plurality of connection lines SL1 to SL4 may be disposed between the second patch antennas 1210 inside the PCB 1010a and the RFIC chip 1400a. A length of each of the plurality of signal connection lines SL1 to SL4 is defined as a connected length between the RFIC 1400a and the phased array antenna portion 1200a.

(138) Each of the plurality of signal connection lines SL1 to SL4 may include a first part SL1, a second part SL2, a third part SL3, and a fourth part SL4. The first part SL1 may be defined on the second surface S2 of the PCB 1010a. The third part SL3 may be formed to configure a coplanar waveguide structure inside the PCB 1010a. The second part SL2 may be defined to electrically connect the first part SL1 and the third part SL3. The plurality of signal connection lines SL1 to SL4 may have the same length.

(139) The antenna module 1000a may include a plurality of ground regions. The antenna module 1000a may include a first ground region 1300g, a second ground region 1200a, and a third ground region 1300g. The two-patch antenna structure and the dummy patches DP11, DP12, DP21, and

DP22 may be disposed in the first ground region **1100g**.

(140) The second ground region **1200g** that is configured as a metal surface may be defined between the coplanar waveguide structure WG2 and the second patch antenna **1210**. The third ground region **1300g** that is configured as a metal surface may be additionally defined between the coplanar waveguide structure WG1 and the RFIC chip **1400a**.

(141) The dummy patches DP11, DP12, DP21, and DP22 which are spaced apart the third gap G3 from the first patch antennas PA11, PA18, PA21, and PA28 may be disposed on the first surface S1 of the PCB **1010a**. The dummy patches DP11, DP12, DP21, and DP22 may also be disposed in the area including the first horizontal ground region GH1, the second horizontal ground region GH2, and the vertical ground walls GV1 to GV4, GV12 to GV34.

(142) The first horizontal ground region GH1 may be formed on the first surface S1. The second horizontal ground region GH2 may be formed in the PCB **1010a**. The second horizontal ground region GH2 may be defined in the sixth layer La6 or the seventh layer La7 of the PCB **1010a**. The vertical ground walls GV1 to GV4, GV12 to GV34 may be formed to vertically connect the first horizontal ground region GH1 and the second horizontal ground region GH2.

(143) The phased array antenna portions **1200a** may include a first array antenna portion **1250a** and a second array antenna portion **1250b**. In the first array antenna portion **1250a**, eight first patch antennas PA11 to PA18 and eight second patch antennas CP11 to CP18 may be disposed in a first column in the Y-axial direction. In the second array antenna portion **1250b**, eight first patch antennas PA21 to PA28 and eight second patch antennas CP21 to CP28 may be disposed in a second column in the Y-axial direction. The phased array antenna portions **1200a** each may be implemented as a 2×8 array antenna.

(144) The first, third, fifth, and seventh antenna elements of the first array antenna portion **1250a** may be disposed such that the second patch antennas CP11, CP13, CP15, and CP17 are offset from the first patch antennas PA11, PA13, PA15, and PA17 in the positive Y-axial direction. The second, fourth, sixth, and eighth antenna elements of the first array antenna portion **1250a** may be disposed such that the second patch antennas CP12, CP14, CP16, and CP18 are offset from the first patch antennas PA12, PA14, PA16, and PA18 in the negative Y-axial direction.

(145) The second patch antennas CP11, CP13, CP15, and CP17 corresponding to the first, third, fifth, and seventh antenna elements may be disposed adjacent to another side (right side) of the ground wall **1130**. The second patch antennas CP12, CP14, CP16, and CP18 corresponding to the second, fourth, sixth, and eighth antenna elements may be disposed adjacent to one side (left side) of the ground wall **1130**.

(146) The first, third, fifth, and seventh antenna elements of the second array antenna portion **1250b** may be disposed such that the second patch antennas CP21, CP23, CP25, and CP27 are offset from the first patch antennas PA21, PA23, PA25, and PA27 in the positive Y-axial direction. The second, fourth, sixth, and eighth antenna elements of the second array antenna portion **1250b** may be disposed such that the second patch antennas CP22, CP24, CP26, and CP28 are offset from the first patch antennas PA22, PA24, PA26, and PA28 in the negative Y-axial direction.

(147) The second patch antennas CP21, CP23, CP25, and CP27 corresponding to the first, third, fifth, and seventh antenna elements may be disposed adjacent to another side (right side) of the ground wall **1130**. The second patch antennas CP22, CP24, CP26, and CP28 corresponding to the second, fourth, sixth, and eighth antenna elements may be disposed adjacent to one side (left side) of the ground wall **1130**.

(148) So far, an electronic device having an antenna module has been described. The technical effects of the electronic device having the antenna module according to the present disclosure are as follows.

(149) An electronic device according to an embodiment of the present disclosure may perform wireless communication of A/V data regardless of the location of an A/V transmitting device through first and second antenna structures in which a plurality of array antennas are disposed.

(150) Furthermore, the A/V transmitting device may transmit two streams of data, thereby minimizing video quality deterioration that occurs when increasing a data compression rate.

(151) In addition, since a horizontally polarized antenna and a vertically polarized antenna can be disposed together on one substrate, thereby allowing an antenna module to be compact and providing a high data reception rate.

(152) Moreover, horizontally and vertically polarized signals may be used according to an array antenna disposition structure of the A/V transmitting device and the electronic device, thereby performing A/V wireless communication with reduced mutual interference while increasing a communication capacity.

(153) Besides, horizontally and vertically polarized signals may be used in consideration of the location of the A/V transmitting device and electronic device the polarization characteristics of the array antennas, thereby performing A/V wireless communication with reduced mutual interference while increasing a communication capacity.

(154) In addition, even when an obstacle is disposed on a wireless communication path between the A/V transmitting device and the electronic device, a beamforming direction may be changed and reflected waves may be used, thereby providing seamless A/V wireless communication.

(155) Also, the number of array antennas disposed in a front area of the antenna module of the A/V transmitting device may be greater than the number of antennas in a side region or bottom region. Accordingly, signals can be transmitted over a longer distance in the front area of the antenna module than in the side region or bottom region. Also, an antenna module that has two-dimensional array antennas and is capable of transmitting signals even upward through beamforming can be implemented.

(156) Also, the number of array antennas disposed in side regions of the antenna module of the A/V transmitting device may be greater than the number of antennas in other areas. Accordingly, an antenna module capable of achieving a wider beam coverage in the side regions than that in a front or bottom region can be implemented.

(157) Further scope of applicability of the present disclosure will become apparent from the following detailed description. It should be understood, however, that the detailed description and specific examples, such as the preferred embodiments of the present disclosure, are given by way of illustration only, since various modifications and alternations within the spirit and scope of the disclosure will be apparent to those skilled in the art. Therefore, the detailed description should not be limitedly construed in all of the aspects, and should be understood to be illustrative. Therefore, all changes and modifications that fall within the metes and bounds of the claims, or equivalents of such metes and bounds are therefore intended to be embraced by the appended claims.

Claims

1. An antenna module implemented as a multi-layered package, the antenna module comprising: a printed circuit board (PCB) having a plurality of layers; an array antenna portion including a plurality of antenna elements disposed on the PCB, wherein each of the plurality of antenna elements has a two-patch antenna structure, first patch antennas of the two-patch antenna structure being located on a first surface of the PCB, the first surface being an outermost surface of the PCB, wherein second patch antennas of the two-patch antenna structure are disposed inside the PCB, and a part of the first patch antenna and a part of the second patch antenna are stacked to overlap each other; a radio frequency integrated circuit (RFIC) chip bonded to a second surface of the PCB, the second surface being another outermost surface of the PCB; and a plurality of signal connection lines configured to connect from the RFIC chip to the array antenna portion, the plurality of signal connection lines are connected and fed respectively to the second patch antennas of the two-patch antenna structure disposed inside the PCB, the first patch antennas disposed on the first surface of the PCB are spaced apart from each other at a third gap, an area between the plurality of second

patch antennas disposed inside the PCB includes a first region in which the second patch antennas are disposed to be spaced apart by a first gap, and a second region in which the second patch antennas are disposed to be spaced apart by a second gap, and a length of the first gap is shorter than a length of the third gap, and the length of the third gap is shorter than a length of the second gap.

2. The antenna module of claim 1, wherein the two-patch antenna structure is disposed in an area including, a first horizontal ground region defined in the first surface, a second horizontal ground region defined inside the PCB, and vertical ground walls vertically connecting the first horizontal ground region and the second horizontal ground region.

3. The antenna module of claim 2, wherein each of the plurality of signal connection lines is disposed between the second patch antennas inside the PCB and the RFIC chip, and each of the plurality of signal connection lines comprises: a first part on the second surface of the PCB; a third part configuring a coplanar waveguide structure inside the PCB; a second part electrically connecting the first part and the third part; and a fourth part electrically connecting the third part and one of the second patch antennas, and the plurality of signal connection lines have the same length.

4. The antenna module of claim 3, wherein a second ground region is defined between the coplanar waveguide structure and the second patch antennas, and wherein a third ground region configured as a metal surface is further defined between the coplanar waveguide structure and the RFIC chip.

5. The antenna module of claim 4, wherein a dummy patch spaced a third gap apart from an outermost first patch antenna among the first patch antennas is disposed on the first surface of the PCB, and the dummy patch includes: the first horizontal ground region defined on the first surface; the second horizontal ground region defined inside the PCB, and the vertical ground walls vertically connecting the first horizontal ground region and the second horizontal ground region.

6. The antenna module of claim 3, wherein the array antenna portion comprises: a first array antenna portion in which eight first patch antennas and second patch antennas are disposed in a first column in a Y-axial direction; and a second array antenna portion in which eight first patch antennas and eight second patch antennas are disposed in a second column in the Y-axial direction, and the array antenna portion is implemented as a 2×8 array antenna.

7. The antenna module of claim 6, wherein first, third, fifth, and seventh antenna elements of the first array antenna portion are disposed such that the second patch antennas are offset from the first patch antennas in a positive Y-axial direction, second, fourth, sixth, and eighth antenna elements of the first array antenna portion are disposed such that the second patch antennas are offset from the first patch antennas in a negative Y-axial direction, first, third, fifth, and seventh antenna elements of the second array antenna portion are disposed such that the second patch antennas are offset from the first patch antennas in the positive Y-axial direction, second, fourth, sixth, and eighth antenna elements of the second array antenna portion are disposed such that the second patch antennas are offset from the first patch antennas in the negative Y-axial direction, the second patch antennas corresponding to the first, third, fifth, and seventh antenna elements are disposed adjacent to another side of the vertical ground walls, and the second patch antennas corresponding to the second, fourth, sixth, and eighth antenna elements are disposed adjacent to one side of the vertical ground walls.

8. The antenna module of claim 7, wherein a first feeding point of the first antenna element is spaced the first gap apart from a second feeding point of the second antenna element, the second feeding point of the second antenna element is spaced a second gap apart from a third feeding point of the third antenna element, the first feeding point connected to a second patch antenna of the first antenna element is offset from a center of the second patch antenna in the positive Y-axial direction, the second feeding point connected to a second patch antenna of the second antenna element is offset from the center of the second patch antenna in the negative Y-axial direction, and the third feeding point connected to a second patch antenna of the third antenna element is offset

from the center of the second patch antenna in the positive Y-axial direction.

9. The antenna module of claim 6, wherein the RFIC chip is disposed on a first layer, and first to fourth ground layers are disposed on second, fourth, fifth, and seventh layers, the first patch antennas and the second patch antennas are disposed on a twelfth layer and a tenth layer of the PCB, respectively, and the fourth part vertically connects the first part of the signal connection lines of a sixth layer and the second patch antennas of the tenth layer.

10. The antenna module of claim 9, wherein first vertical vias corresponding to the second part are disposed on a first side surface as an upper area of the RFIC chip, a second side surface as one side area, and a fourth side surface as another side area, and second vertical vias corresponding to the fourth part includes a pair of via groups that are spaced apart from each other partially by the first gap and partially by the second gap.

11. The antenna module of claim 10, wherein first, third, fifth, and seventh antenna elements of the first array antenna portion are disposed such that connection points of the second vertical vias are offset from the first patch antennas in the positive Y-axial direction, second, fourth, sixth, and eighth antenna elements of the first array antenna portion are disposed such that connection points of the second vertical vias are offset from the first patch antennas in the negative Y-axial direction, first, third, fifth, and seventh antenna elements of the second array antenna portion are disposed such that connection points of the second vertical vias are offset from the first patch antennas in the positive Y-axial direction, second, fourth, sixth, and eighth antenna elements of the second array antenna portion are disposed such that connection points of the second vertical vias are offset from the first patch antennas in the negative Y-axial direction, the second vertical vias corresponding to the first, third, fifth, and seventh antenna elements are disposed adjacent to another side of the vertical ground walls, and the second vertical vias corresponding to the second, fourth, sixth, and eighth antenna elements are disposed adjacent to one side of the vertical ground walls.
